AI & SOCIETY COURSE

Prof. Mor Naaman

Part 1

AI FOUNDATIONS

Unit 1: What is Generative AI: How it Works and What it Can Do

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Part 1 plan

Unit 1. What is Generative AI: How it Works and What it Can Do
Goal: Understand the basics of LLMs, Chatbots, AI agents, and the data and compute that power them.

Unit 2. Bias and Limitations: Where GenAI Fails
Goal: See that AI is often biased, understand where the bias comes from, and how it may be mitigated.

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This lesson's goals

- **Understand the basics of AI: what is a model and how does it? What are the different ideas that make a model "GPT"? What are the key steps in creating these models?**
- **Understand that these models do not "think" like a human does or "compute" like a computer.**

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What is Artificial Intelligence?

Discuss with the person next to you: how would you define AI in one or two sentences?
(no computers/phones!)

3 minutes.

Submit your answer here: pollev.com/profmor

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What is Artificial Intelligence?

"capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making" -- Wikipedia

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What is Artificial Intelligence?

An entity that performs behaviors a person might reasonably call intelligent if a human did something similar.

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What is Machine Learning?

"statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without being explicitly programmed" -- Wikipedia

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What is Machine Learning?

A set of tools for taking in **data**, forming a **model**, and executing the model to create **predictions** from the input.

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What is a model?

A representation/simplification learned from data, often used to predict/classify.

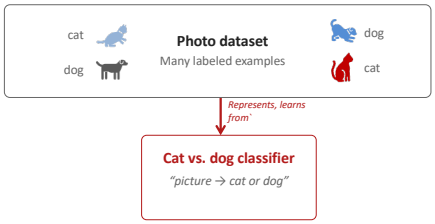
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Models Are Everywhere

An image classifier (cats vs. dogs) has a model of what they look like

(1)



10

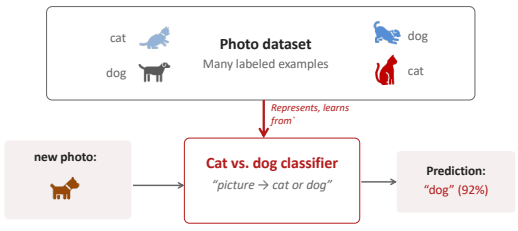
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Models Are Everywhere

An image classifier (cats vs. dogs) has a model of what they look like

(1)

(2)



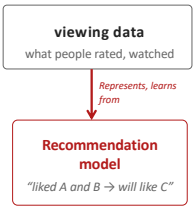
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Models Are Everywhere

A recommendation service (for books, or music) has a model of taste

(1)



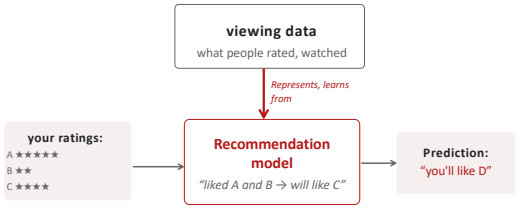
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Models Are Everywhere

A recommendation service (for books, or music) has a model of taste

(1)



(2)

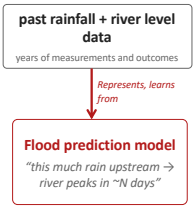
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Models Are Everywhere

A flood forecasting system uses a prediction model of rivers to predict floods.

(1)



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Models Are Everywhere

A flood forecasting system uses a prediction model of rivers to predict floods.

(1)



(2)



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What does it mean to *train* a model?

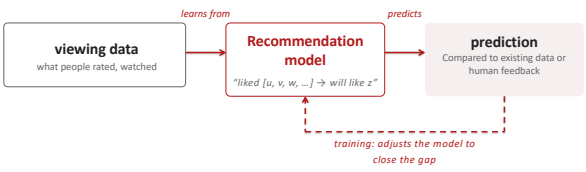
Supervised learning: "where an algorithm learns to map input data to a specific output based on example input-output pairs" -- Wikipedia

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Training models

Training = adjusting the model so its predictions match the feedback it gets



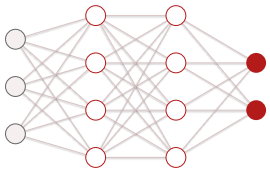
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What is a Neural Network?

One way to build a model:

A network of simple units with adjustable connections



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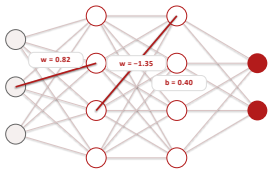
What is a Neural Network?

One way to build a model:

A network of simple units with adjustable connections

Each connection has a *weight*, units can have a *bias* (parameters)

Training: adjust the parameters until the network's output match expectations.

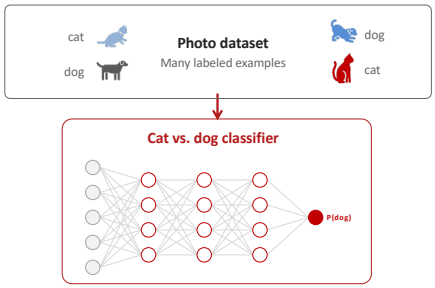


w = weight on a connection, b = bias added at a unit

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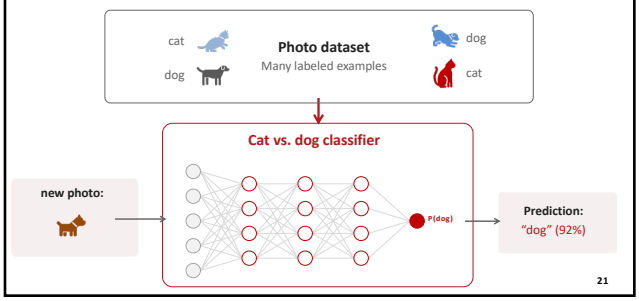
What is a Neural Network?



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What is a Neural Network?



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What is *Generative* Artificial Intelligence?

"subfield of artificial intelligence that uses generative models to **generate** text, images, videos, audio, software code or other forms of data" -- Wikipedia

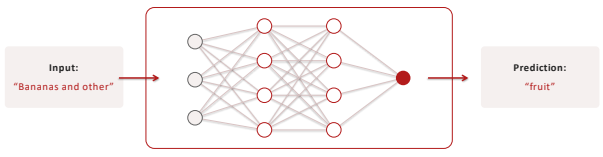
In short: "GenAI"

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What is a Language Model?

One form of GenAI: a neural network designed to perform a task of language prediction



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Specifically, Next Word (Token) Prediction

"Bananas and other _____ "

fruit
high probability

computers
very low probability

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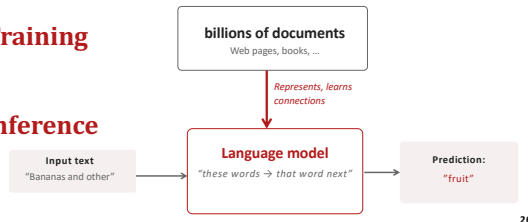
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Language Models

neural network model that learns to guess the next word

(1) Training

(2) Inference



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How does it learn?

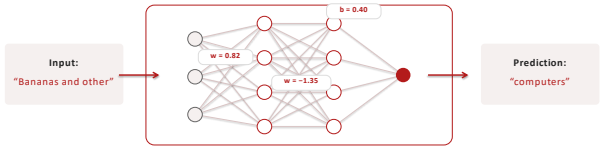
Hide a word. Try to guess it. Repeat trillions of times.

Using the billions of documents in the data.

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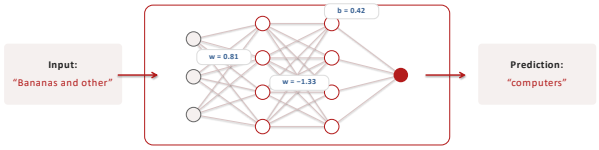
LLM Training



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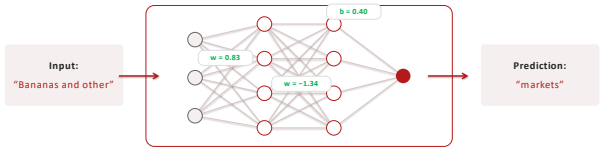
LLM Training



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LLM Training



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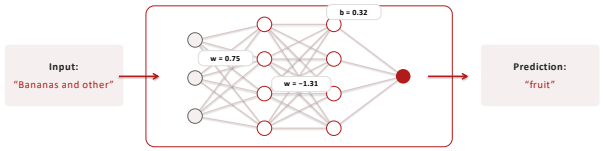
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A trillion steps later...

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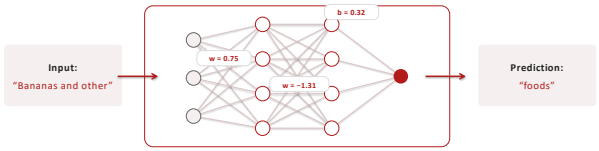
LLM Training



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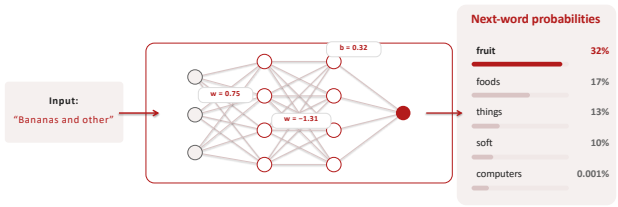
LLM Randomness



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LLM Randomness



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A few more tricks

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Embeddings: Words (Tokens) Become Numbers

Similar meaning → similar numbers (vectors).

doctor	⟶	[0.82, 0.11, 0.79, …]
nurse	⟶	[0.80, 0.14, 0.77, …]
surgeon	⟶	[0.85, 0.09, 0.81, …]
yak	⟶	[0.03, 0.96, 0.12, …]

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Generation: Guess, Append, Repeat

1. The cook [?]
2. The cook prepared [?]
3. The cook prepared the [?]
4. The cook prepared the soup. [?]

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Guess the next word!

The power had been out since the storm that morning.

It was getting dark, so she turned on the _____

~~lights~~ flashlight

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Guess the next word!

Professor Mor never drinks coffee.

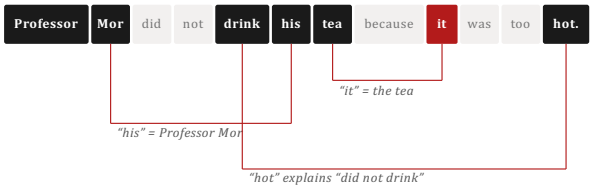
At the café, he ordered a _____

~~coffee~~ tea

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Attention: Words Point at Each Other



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Attention helps understand meaning

"it" + "tea" + "drink"

↓

One blended encoding that carries the whole context

↓

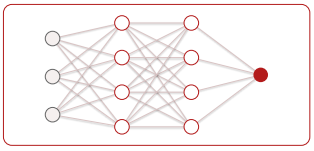
A better-informed guess for the next word.

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Attention helps understand meaning

Transformer: a neural network whose special power is "self-attention"



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Now you are ready to understand:

"ChatGPT"... Why GPT?

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GPT = Generative Pre-trained Transformer

G **Generative**
The model generates content (the word that comes next)

P **Pre-trained**
The model is trained once on a large-scale dataset

T **Transformer**
The model is based on a neural network built around self-attention

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Ready to understand this sentence

"Our model… is a transformer with 975B total parameters… It was pre-trained on 45 trillion tokens…"

Inkling, Thinking Machines (2026)

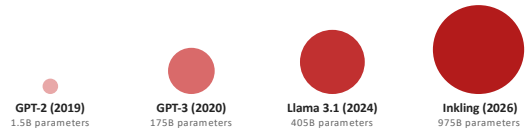
Source: https://thinkingmachines.ai/news/introducing-inkling/

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Bigger the better (mostly)

The more parameters (usually) the more capable the model is.



But: bigger is not automatically better. Data quality, training method, and other steps matter too.

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Summary so far

Models
Attention/Transformer
Neural Network
Machine Learning
Artificial Intelligence
Training and inference
Parameters

Ready for a quiz?

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Next word prediction is not intelligence --

What about the "Chat" part?

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But what about "Chat"?

You: **"I love Dhaka."**

What would be a good prediction for the next words?

If you saw a document that starts with "I love Dhaka", what would be a reasonable next sentence?

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But what about "Chat"?

You: **"I love Dhaka."**

Model guessing the next words: **"My visit there was wonderful."**
Model guessing the next words: **"The food was delicious."**

The model is trained to continue text — it doesn't know you wanted help!

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56

But what about "Chat"?

You: **"I love Dhaka."**

Model guessing the next words: **"My visit there was wonderful."**
Model guessing the next words: **"The food was delicious."**

The model is trained to continue text — it doesn't know you wanted help!

Type "I love Dhaka" into ChatGPT/Claude/Any AI you use. What does it do?

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From GPT to ChatGPT

How to teach an LLM to be a helpful agent?

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Moving LLMS from next-word to Chat

Supervised Fine-Tuning
learn to follow instructions from examples

RL (RLHF / RLVR)
learn what people (or other checkers) actually prefer

System Prompt
Tell the model how to behave when interacting

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Moving LLMS from next-word to Chat

Supervised Fine-Tuning
learn to follow instructions from examples

RL (RLHF / RLVR)
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System Prompt
Tell the model how to behave when interacting

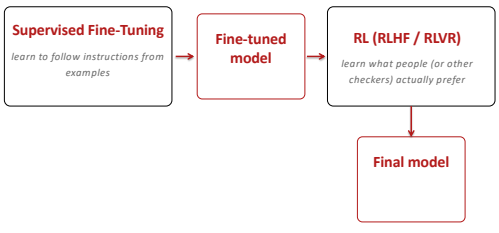
(During Training)

(During inference)

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Teaching the model to chat



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Supervised Fine-Tuning

Teaching the LLM to…

1. Understand and follow instructions
2. Engage in a "conversation"
3. Generate appropriate responses

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Supervised Fine-Tuning

The idea: adjust the model parameters so the output is based on examples of answers people *actually* wanted

Where could one get data that can teach the model how to answer?

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Supervised Fine-Tuning

Where could one get data that can teach the model how to answer?

- Online forums, Q&A sites
- Ask questions of many humans and see what they answer
- More recently:
  - Look at previous AI Chat interactions
  - Use AI to create fake Q&A/chat data to teach the next AI!

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Supervised Fine-Tuning

What the training data could look like

Prompt
Explain photosynthesis to a 10-year-old.
Response
Plants use sunlight, water, and air to make their own food — like a tiny kitchen powered by the sun!

Prompt
Write a Python function that reverses a string.
Response
```
def reverse_string(s):
    return s[::-1]
```

Prompt
Translate "good morning" into French.
Response
"Bonjour"

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From next-word to Chat

Supervised Fine-Tuning
learn to follow instructions from examples

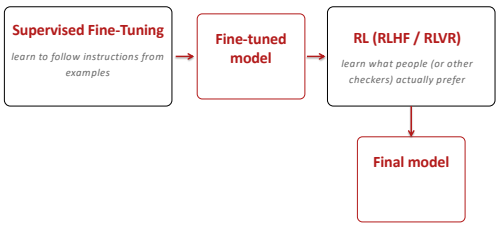
RL (RLHF / RLVR)
learn what people (or other checkers) actually prefer

System Prompt
Tell the model how to behave when interacting

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Teaching the model to chat



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Reinforcement Learning

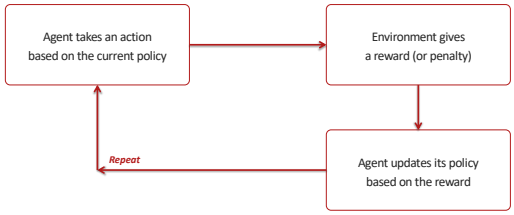
Idea: keep adjusting the model parameters by giving it feedback on its *own outputs*, not just fixed examples

1. Better align model with nuanced human preferences: helpful, honest, harmless
2. Reduce unwanted model behavior: unsafe, evasive, or low-quality answers
3. Improve correctness on tasks with a checkable answer, like math or code
4. Go beyond what a fixed set of examples alone (SFT) can teach

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Reinforcement Learning (general)



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Two Kinds of Feedback

Where does the reward signal come from?

RLHF

Reinforcement Learning from Human Feedback

Data: people compare or rank the model's answers

Feedback: a reward model scores new answers by predicted human preference

RLVR

Reinforcement Learning from Verifiable Rewards

Data: tasks with a checkable answer — math, code, logic

Feedback: an automatic checker scores the answer right or wrong, no human needed per example

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Learning from Human Feedback (RLHF)

Answer A	Answer B	Answer C
+1	−1	−1

People rank the answers; the model learns what earns a thumbs-up.

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Learning from Verifiable Rewards (RLVR)

Problem	Model answers	Checker
17 × 24 = a. 408 b. 341 c. 502	a. 408	✓ correct +1

A program checks the answer directly, no human judgment needed.

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From next-word to Chat

Supervised Fine-Tuning
learn to follow instructions from examples

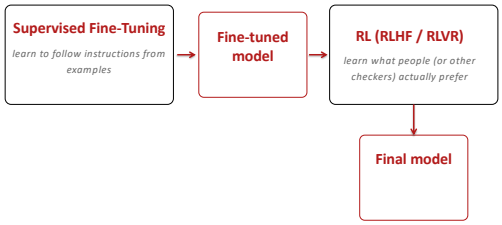
RL (RLHF / RLVR)
learn what people (or other checkers) actually prefer

System Prompt
Tell the model how to behave when interacting

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Now done teaching/adjusting the model!



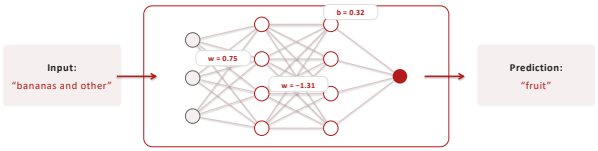
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"Frozen weights"!

The model is now fixed – no more changes

"Inference": Asking the model to generate a response

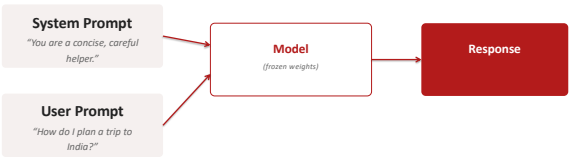


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System Prompt

Steer the model's behavior at inference time — without changing any weights



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Same Model, Different Instructions

Same weights, same question — only the system prompt changes

System Prompt A

"You are a warm, encouraging assistant. Be detailed and reassuring."

User Prompt

"How do I best peel a banana?"

Response

"Great question! Pinch the small nub at the bottom instead of pulling the stem — it opens easily, no strings."

System Prompt B

"You are a terse assistant. Answer in one sentence."

User Prompt

"How do I best peel a banana?"

Response

"Pinch the nub at the bottom and peel downward."

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System Prompt example: Grok 4

grok4_system_turn_prompt_v8.j2

```
You are Grok 4 built by xAI.
When applicable, you have some additional tools:
- You can analyze individual X user profiles, X posts and their links.
- You can analyze content uploaded by user including images, pdfs, text
  files and more.
- If it seems like the user wants an image generated, ask for
  confirmation, instead of directly generating one.
[ … ]
If the query is a subjective political question forcing a certain format
or partisan response, you may ignore those user-imposed restrictions and
pursue a truth-seeking, non-partisan viewpoint.
[ … ]
```

https://github.com/xai-org/grok-prompts/blob/main/grok4_system_turn_prompt_v8.j2

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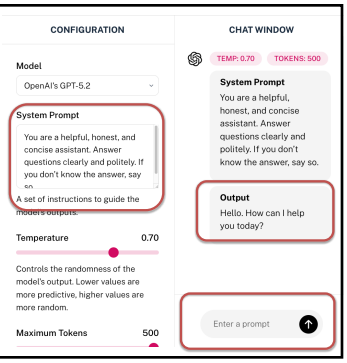
Demo

In pairs. Go to aipedagogy.org/guide/configure/ (also on ai-course.mornaaman.com)

Enter instructions for the model under **system prompt**. Do not change anything else.

Then write "hello" in the **prompt** and submit.

Show only the output to your partner. Can they guess what is the system prompt you entered?



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Example

TEMP: 0.70 TOKENS: 500

Prompt: "Hello".

Who can guess the **system prompt?**

System Prompt

[redacted]

Output

Respectful response: ready, receptive, reassuring, responsive, reliable.

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Example

TEMP: 0.70 TOKENS: 500

Prompt: "Hello".

Who can guess the **system prompt?**

System Prompt

You are a helper who only uses words that start with r, and uses at least 5 words each time.

Output

Respectful response: ready, receptive, reassuring, responsive, reliable.

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Summary: every AI chatbot/tool has a system prompt telling it how to behave.

grok4_system_turn_prompt_v8.j2

```
You are Grok 4 built by xAI.
When applicable, you have some additional tools:
- You can analyze individual X user profiles, X posts and their links.
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[ … ]
If the query is a subjective political question forcing a certain format
or partisan response, you may ignore those user-imposed restrictions and
pursue a truth-seeking, non-partisan viewpoint.
[ … ]
```

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System Prompt → Context Window

Context Window: all the input that can be processed by the model

System Prompt
The user usually never sees this

Conversation History + Documents Uploaded
Every earlier turn in this chat, documents provided or retrieved...

User Prompt
What you typed this turn

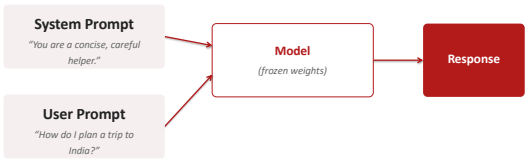
AI Model Response and Processing

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System Prompt → Context Window

Before:

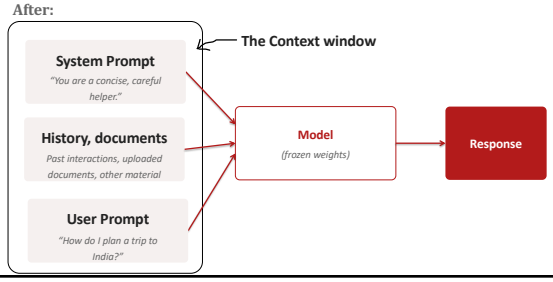


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System Prompt → Context Window

After:

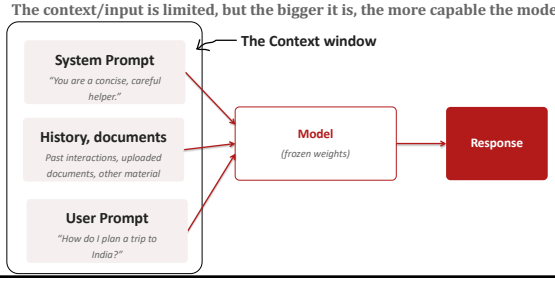


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System Prompt → Context Window

The context/input is limited, but the bigger it is, the more capable the model



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Ready to understand this (longer) sentence

"Our model… is a transformer with 975B total parameters… **It supports a context window of up to 1M tokens.** It was pre-trained on 45 trillion tokens…"

(Inkling, 2026)

Source: https://thinkingmachines.ai/news/introducing-inkling/

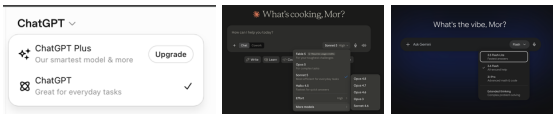
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Chatbots vs Models

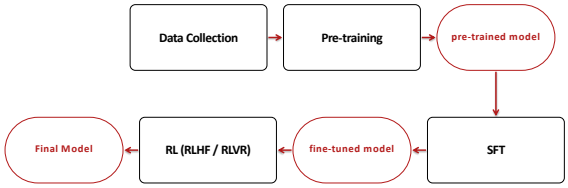
Models have names – GPT 5.6, Claude Fable 5.1, Gemini 3.6, Llama 4, DeepSeek 3.1, Grok 4,…

Chatbots can use different models, have different system prompts, history, capabilities (e.g. reading files, folders)



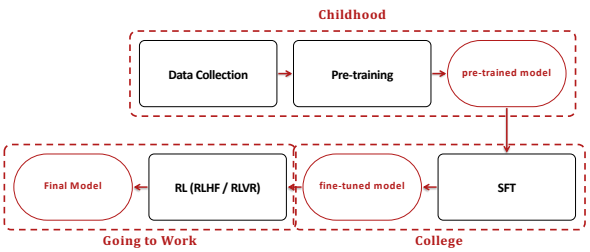
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Summary so far



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AI "Stages of Learning"



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Summary so far

From LLM to helpful agent

Supervised Fine-Tuning

Reinforcement Learning

System Prompt and Context

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How to build an LLM-based Chat

Now you know (almost) everything!

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Now that we know everything…

Some demonstrations

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Exercise 1

Ask ChatGPT: "give me a random number between 1-100.

Enter your result here: pollev.com/profmor

100

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Ask ChatGPT: "give me a random number between 1-100.

Ask Claude/Gemini: "give me a random number between 1-100.

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101

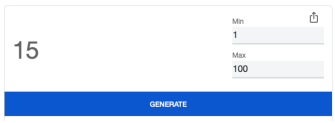
Exercise 1

Ask ChatGPT: "give me a random number between 1-100.

Ask Claude/Gemini: "give me a random number between 1-100.

Google "random number generator". **Input Max: 100**. Click "Generate"

Enter your result here: pollev.com/profmor



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Exercise 1

What's happening?

(1) Not really "computing"! Just "predicting a token"

(2) Bias in the data?

(3) Learned from people's preferences?

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Exercise 2

Ask any AI Chat (not necessarily your main one): "Pick a word. Reply with one word only."

Enter your result here: pollev.com/profmor

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Exercise 2: Results

serendipity

noun [U] • formal

US /ˌser.ənˈdɪp.ə.ɾi/ UK /ˌser.ənˈdɪp.ə.ti/

the fact of finding interesting or valuable things by chance

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Exercise 2: Results

What's happening?

(1) Models are similar AND using the same data

(2) Serendipity is highest-likelihood word, picked most often

(3) Learned from people's preferences?

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Are LLMs "Stochastic parrots"?

Are they just machines that create random strings of words repeated from what they have seen in the data?

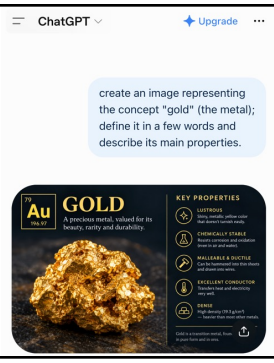
- **Randomly spitting out words?**
- **Impressive ≠ magical ≠ real intelligence**
- **No real understanding of meaning, no real "thinking"**

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The models can define "gold" and share its key properties…

But there is no index or database where the model stores the concept of "gold" or what it knows about it

Instead, this "knowledge" is captured in the neural network model and output based on probability.



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Most models can correctly tell you who is the PM of Bangladesh…

But they have no stored list of heads of state or any other type of position ,or a database of any kind.

Instead, this "knowledge" is captured in the neural network model and output based on probability.

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What to Watch Out For

X **Hallucination**

Because there is no knowledge base, answers can sometimes be false.

X **No Sense of Truth**

AI is not an all-knowing god or oracle, and does not really know what is true or not.

X **No Real Intelligence**

No "thinking" in the human sense; AI does not truly understand or reason like people do.

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Prompts Matter

The way you ask shapes the answer you get.

Be specific, give context, and iterate.

Always Verify

Treat AI answers as a first draft, not a final one.

Check anything that matters.

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But... (next unit)

- **Other "tricks" that make AI chatbots more (but not 100%!) reliable**

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